Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

**IT13: ADBMS**

## Practical Assignment 2 Solution Template

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**Practical 2: Implementation of different types of functions with suitable examples.**

- Number Function
- Aggregate Function
- Character Function
- Conversion Function
- Date Function

**LAB PRACTICE ASSIGNMENT:**

Create a table EMPLOYEE with the following schema:

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name,Job_id, Designation, Salary)

Write SQL statements for the following queries.

Solution: -

```
Create Table EMPLOYEE (
EMP_no INT PRIMARY KEY,
E_name VARCHAR (50),
E_address VARCHAR (100),
e_ph_no VARCHAR (15),
Dept_no INT,
Dept_name VARCHAR (50),
Job_id INT,
Designation VARCHAR (30),
Salary VARCHAR (12),
Experience INT,
Join_date date
```

);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| EMP_no | int | NO | PRI | NULL | |
| E_name | varchar(50) | YES | | NULL | |
| E_address | varchar(100) | YES | | NULL | |
| e_ph_no | varchar(15) | YES | | NULL | |
| Dept_no | int | YES | | NULL | |
| Dept_name | varchar(50) | YES | | NULL | |
| Job_id | int | YES | | NULL | |
| Designation | varchar(30) | YES | | NULL | |
| Salary | decimal(10,2) | YES | | NULL | |
| Experience | int | YES | | NULL | |
| Join_date | date | YES | | NULL | |

1. List the E_no, E_name, Salary of all employees working for MANAGER.

Solution: -

```
SELECT EMP_no, E_name, salary
FROM EMPLOYEE
WHERE Designation = 'MANAGER';
```

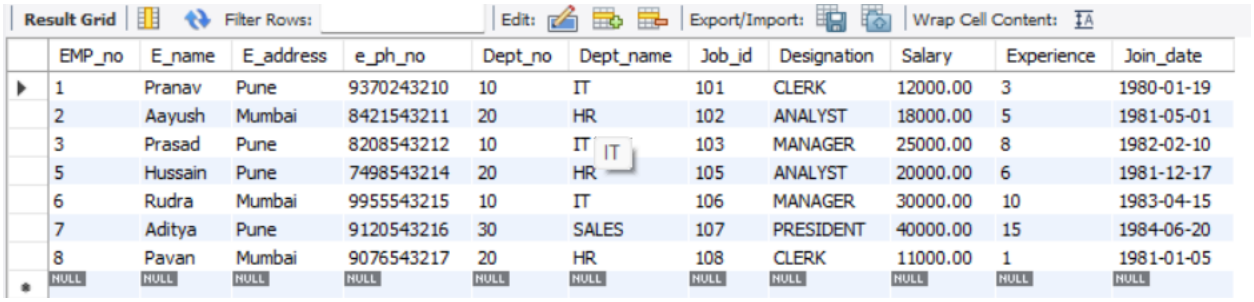
Result Grid | Filter Rows: | Edit:

| EMP_no | E_name | salary |
|---|---|---|
| 3 | Prasad | 25000.00 |
| 6 | Rudra | 30000.00 |
| NULL | NULL | NULL |

2. Display all the details of the employee whose salary is more than 10000.

Solution: -

```
SELECT *
FROM EMPLOYEE
WHERE Salary > 10000;
```

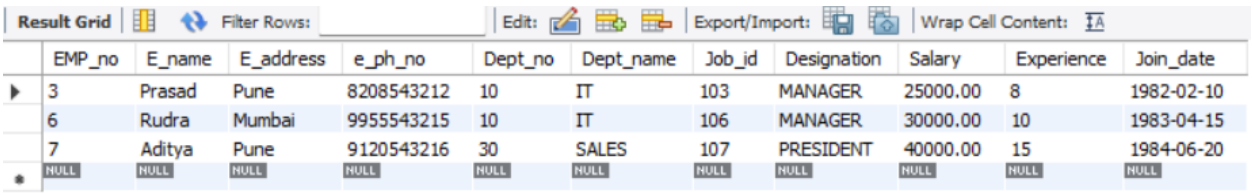


| EMP_no | E_name | E_address | e_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Experience | Join_date |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Pranav | Pune | 9370243210 | 10 | IT | 101 | CLERK | 12000.00 | 3 | 1980-01-19 |
| 2 | Aayush | Mumbai | 8421543211 | 20 | HR | 102 | ANALYST | 18000.00 | 5 | 1981-05-01 |
| 3 | Prasad | Pune | 8208543212 | 10 | IT | 103 | MANAGER | 25000.00 | 8 | 1982-02-10 |
| 5 | Hussain | Pune | 7498543214 | 20 | HR | 105 | ANALYST | 20000.00 | 6 | 1981-12-17 |
| 6 | Rudra | Mumbai | 9955543215 | 10 | IT | 106 | MANAGER | 30000.00 | 10 | 1983-04-15 |
| 7 | Aditya | Pune | 9120543216 | 30 | SALES | 107 | PRESIDENT | 40000.00 | 15 | 1984-06-20 |
| 8 | Pavan | Mumbai | 9076543217 | 20 | HR | 108 | CLERK | 11000.00 | 1 | 1981-01-05 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

3. List the employees in ascending order of Designations of those who joined after 1981.

Solution: -

```
SELECT*
FROM EMPLOYEE
WHERE Join_date >= '1982-01-01'
ORDER BY DESIGNATION ASC;
```



| EMP_no | E_name | E_address | e_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Experience | Join_date |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | Prasad | Pune | 8208543212 | 10 | IT | 103 | MANAGER | 25000.00 | 8 | 1982-02-10 |
| 6 | Rudra | Mumbai | 9955543215 | 10 | IT | 106 | MANAGER | 30000.00 | 10 | 1983-04-15 |
| 7 | Aditya | Pune | 9120543216 | 30 | SALES | 107 | PRESIDENT | 40000.00 | 15 | 1984-06-20 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

4. List the employees along with their Experience and Daily Salary.

Solution: -

```
SELECT E_name, Experience, Salary , salary / 30 As Daily_Salary
FROM EMPLOYEE;
```



| E_name | Experience | Salary | Daily_Salary |
|---|---|---|---|
| Pranav | 3 | 12000.00 | 400.000000 |
| Aayush | 5 | 18000.00 | 600.000000 |
| Prasad | 8 | 25000.00 | 833.333333 |
| Gaurav | 2 | 9000.00 | 300.000000 |
| Hussain | 6 | 20000.00 | 666.666667 |
| Rudra | 10 | 30000.00 | 1000.000000 |
| Aditya | 15 | 40000.00 | 1333.333333 |
| Pavan | 1 | 11000.00 | 366.666667 |
| Shashank | 3 | 2000.00 | 66.666667 |

5. List the employees who are either 'CLERK' or 'ANALYST '.

Solution: -

```
SELECT *
FROM EMPLOYEE
WHERE DESIGNATION = 'CLERK'
OR DESIGNATION = 'ANALYST';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| EMP_no | E_name | E_address | e_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Experience | Join_date |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Pranav | Pune | 9370243210 | 10 | IT | 101 | CLERK | 12000.00 | 3 | 1980-01-19 |
| 2 | Aayush | Mumbai | 8421543211 | 20 | HR | 102 | ANALYST | 18000.00 | 5 | 1981-05-01 |
| 4 | Gaurav | Nashik | 9022543213 | 30 | SALES | 104 | CLERK | 9000.00 | 2 | 1981-12-03 |
| 5 | Hussain | Pune | 7498543214 | 20 | HR | 105 | ANALYST | 20000.00 | 6 | 1981-12-17 |
| 8 | Pavan | Mumbai | 9076543217 | 20 | HR | 108 | CLERK | 11000.00 | 1 | 1981-01-05 |
| 9 | Shashank | Sangli | 8989502410 | 30 | IT | 109 | CLERK | 2000.00 | 3 | 1981-01-19 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

6. List the employees who joined on 1-MAY-1981, 3-DEC-1981, 17-DEC-1981, 19-JAN-1980.

Solution: -

```
SELECT *
FROM EMPLOYEE
WHERE Join_date IN (
'1981-05-01',
'1981-12-03',
'1981-12-17',
'1980-01-19'
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| EMP_no | E_name | E_address | e_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Experience | Join_date |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Pranav | Pune | 9370243210 | 10 | IT | 101 | CLERK | 12000.00 | 3 | 1980-01-19 |
| 2 | Aayush | Mumbai | 8421543211 | | HR | 102 | ANALYST | 18000.00 | 5 | 1981-05-01 |
| 4 | Gaurav | Nashik | 9022543213 | 30 | SALES | 104 | CLERK | 9000.00 | 2 | 1981-12-03 |
| 5 | Hussain | Pune | 7498543214 | 20 | HR | 105 | ANALYST | 20000.00 | 6 | 1981-12-17 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

7. List the employees who are working for Deptno 10 or20.

Solution: -

```
SELECT *
FROM EMPLOYEE
```

WHERE dept_no IN (10, 20);

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| EMP_no | E_name | E_address | e_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Experience | Join_date |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Pranav | Pune | 9370243210 | 10 | IT | 101 | CLERK | 12000.00 | 3 | 1980-01-19 |
| 2 | Aayush | Mumbai | 8421543211 | 20 | HR | 102 | ANALYST | 18000.00 | 5 | 1981-05-01 |
| 3 | Prasad | Pune | 8208543212 | 10 | IT | 103 | MANAGER | 25000.00 | 8 | 1982-02-10 |
| 5 | Hussain | Pune | 7498543214 | 20 | HR | 105 | ANALYST | 20000.00 | 6 | 1981-12-17 |
| 6 | Rudra | Mumbai | 9955543215 | 10 | IT | 106 | MANAGER | 30000.00 | 10 | 1983-04-15 |
| 8 | Pavan | Mumbai | 9076543217 | 20 | HR | 108 | CLERK | 11000.00 | 1 | 1981-01-05 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

8. List the Enames that start with 'S'.

Solution: -

```
INSERT INTO EMPLOYEE VALUES
(9, 'Shashank', 'Sangli', '8989502410', 30, 'IT', 109, 'CLERK' , 2000 , 3, '1981-01-19');


SELECT E_name
FROM EMPLOYEE
WHERE E_NAME LIKE 'S%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| E_name |
|---|
| Shashank |

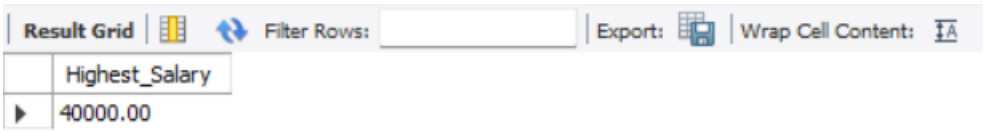
9. Write a SQL query to find the highest salary. Return the highest salary.

Solution: -

```
ALTER TABLE EMPLOYEE
MODIFY COLUMN Salary DECIMAL (10, 2);


SELECT * FROM EMPLOYEE;


SELECT MAX(Salary) AS Highest_Salary
FROM EMPLOYEE;
```




| | Highest_Salary |
|---|---|
| ▶ | 40000.00 |

10. Display the name and the first five characters of the name starting with 'H'. List all the employees except 'PRESIDENT' & 'MGR" in ascending order of Salaries.

Solution: -

```
UPDATE EMPLOYEE
SET E_name = "Hussain"
WHERE EMP_no = 5;


SELECT E_name
FROM EMPLOYEE
WHERE E_name Like 'H%';


SELECT E_name , LEFT(E_name,5) As First_Five
FROM EMPLOYEE
WHERE E_Name LIKE 'H%'
ORDER BY SALARY ASC;


SELECT E_name, substr(E_name,1,5)
FROM EMPLOYEE
WHERE E_name LIKE 'H%';
```




| | E_name | First_Five |
|---|---|---|
| ▶ | Hussain | Hussa |

11. Display the minimum and maximum salary and Experience of the employee.

Solution: -

```
SELECT MIN(Salary) As MIN_Salary,
```

MAX(salary) AS MAX_Salary,

MIN(Experience) AS MIN_Experience,

MAX(Experience) AS Max_Experience

FROM EMPLOYEE;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| MIN_Salary | MAX_Salary | MIN_Experience | Max_Experience |
|---|---|---|---|
| 2000.00 | 40000.00 | 1 | 15 |

12. Display the total salary of all employees.

Solution: -

SELECT SUM(Salary)AS Total_Salary

FROM EMPLOYEE;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Total_Salary |
|---|
| 167000.00 |

13. Write a query to display the average salary of employees working in IT department.

Solution: -

SELECT AVG(Salary) AS AVG_Salary

FROM EMPLOYEE

Where Dept_name = 'IT';

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| AVG_Salary |
|---|
| 17250.000000 |

14. Display the Distinct salary of an employee.

Solution: -

SELECT distinct SALARY

FROM EMPLOYEE;

Result Grid | Filter Rows:

| SALARY |
|---|
| 12000.00 |
| 18000.00 |
| 25000.00 |
| 9000.00 |
| 20000.00 |
| 30000.00 |
| 40000.00 |
| 11000.00 |
| 2000.00 |

15. Drop the database.

Solution: -

```
DROP DATABASE EmployeeDB;
```

57 15:22:37 DROP DATABASE Company 1 row(s) affected 0.094 sec